CERTIFIED CIGAR SOMMELIER (CCS®)

MODULE 5 TEACHING MANUAL

CAP-FOOT DIAGNOSTICS & FLAVOUR TIMING

Status: Core Module – Mandatory
Applies to: CCS® Level I & ACS® Level II
Estimated Study Load: 3 Hours

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MODULE PURPOSE

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This module teaches candidates how to:

- measure and interpret moisture differences between the cap and foot of a cigar,
- predict flavour timing before lighting,
- diagnose imbalance patterns,
- and manage cigar performance professionally.

Many inconsistent smoking experiences are caused not by:

- poor blending,
- factory defects,
- or brand quality,

but by uneven moisture distribution inside the cigar.

This module introduces one of the most powerful diagnostic tools in professional cigar service:

Cap-Foot Delta Analysis.

By mastering this framework, the sommelier can:

- predict whether a cigar will "open late,"
- anticipate harshness or muddiness,
- identify equilibration problems,
- and intentionally shape the customer experience.

This is predictive diagnostics rather than subjective opinion.

LEARNING OUTCOMES

By the end of this module, the candidate shall be able to:

1. Explain why cigars equalize unevenly along their length
2. Measure cap and foot RH correctly
3. Interpret cap-foot deltas accurately
4. Predict flavour timing across the smoke
5. Diagnose common imbalance patterns
6. Recommend corrective actions professionally
7. Explain diagnostic findings clearly to customers

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SECTION 1

WHY AXIAL EQUALIZATION MATTERS

Why This Matters Professionally

Many smokers assume a cigar humidifies uniformly.

In reality, moisture moves through a cigar unevenly and slowly.

This uneven movement creates:

- flavour timing shifts,
- combustion instability,
- and inconsistent smoking behaviour.

Understanding axial equalization allows the sommelier to:

- predict smoking performance before ignition,
- identify preparation problems,
- and improve service consistency.

1.1 The Cigar as an Asymmetric System

A cigar is not structurally symmetrical.

Different parts of the cigar exchange moisture at different speeds.

Foot

- Open structure
- Highly exposed
- Fast moisture exchange

Body

- More insulated
- Slower diffusion

Cap

- Sealed and layered

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- Slowest moisture exchange area

As a result:

- moisture gradients develop naturally,
- equalization takes time,
- and different sections of the cigar combust differently during the smoke.

1.2 Why This Affects Flavour Timing

A cigar burns progressively from:

foot → body → cap.

If moisture levels vary along that axis:

- combustion temperature changes during the smoke,
- flavour release timing shifts,
- and sweetness, strength, and aroma may appear too early or too late.

This means flavour progression can often be predicted before lighting the cigar.

SECTION 2

MEASURING CAP AND FOOT RH

2.1 Measurement Objectives

Cap-foot measurement is used to:

- estimate internal moisture distribution,
- identify equilibration status,
- and detect axial imbalance.

It is not intended to replace:

- long-term humidor management,
- or proper storage procedures.

2.2 Professional Measurement Technique

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Standard Procedure

1. Use a calibrated leaf-level RH probe
(Example: CigarMedics)
2. Measure:
    - the foot,
    - and the cap area through the wrapper
3. Hold the probe until readings stabilize
4. Record:
    - RH values,
    - timestamp,
    - and environmental conditions

Consistency is critical for meaningful diagnostics.

2.3 Common Measurement Errors

Error	Consequence
Measuring immediately after removal from humidor	False RH gradients
Measuring only the foot	Misses core behaviour
Ignoring temperature	Misinterpreted RH values
Inconsistent probe depth	Unreliable comparisons

Professional Perspective

Good diagnostics require:

- repeatability,
- patience,
- and controlled measurement conditions.

Poor measurement technique creates misleading conclusions.

SECTION 3

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INTERPRETING CAP-FOOT DELTAS

3.1 Defining Delta

Cap-Foot Delta is the absolute difference between:

- cap RH,
  and
- foot RH.

This value is one of the strongest predictors of:

- cigar readiness,
- flavour timing,
- and combustion balance.

3.2 Delta Interpretation Table

Delta	Interpretation	Smoking Prediction
≤ 1.0%	Fully equalized	Balanced from first light
1.0–1.5%	Acceptable	Minor flavour timing shift
1.5–2.0%	Transitional	Delayed or uneven flavour
> 2.0%	Unequalized	Significant imbalance likely

3.3 Why Delta Direction Matters

The direction of the imbalance matters just as much as the size of the delta.

A cigar where:

- Foot > Cap
  behaves very differently from one where:
- Cap > Foot.

The sommelier must evaluate:

- both magnitude,
- and direction.

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SECTION 4

FOOT > CAP (UNDER-HUMIDIFIED CORE)

4.1 Typical Readings

Example:

- Foot: 62%
- Cap: 59%

4.2 Predicted Smoking Behaviour

Early Smoke

- brighter flavours,
- sharper spice,
- stronger wrapper influence.

Middle Third

- strength spikes,
- sweetness arrives late,
- pepper intensifies.

Final Third

- heat buildup,
- bitterness risk,
- combustion aggression.

What the Smoker Will Notice

The cigar often feels:

- unbalanced early,
- stronger than expected,
- and increasingly aggressive.

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4.3 Root Cause

Typical causes include:

- recent RH increase,
- insufficient equilibration time,
- dense filler structure resisting moisture absorption.

4.4 Corrective Actions

Recommended approach:

- allow additional resting time,
- maintain stable target RH,
- avoid aggressive dry-boxing,
- and recheck after 48–72 hours.

SECTION 5

CAP > FOOT (OVER-HUMIDIFIED CORE)

5.1 Typical Readings

Example:

- Foot: 60%
- Cap: 63%

5.2 Predicted Smoking Behaviour

Early Smoke

- creamy texture,
- muted flavour,
- slower ignition.

Middle Third

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- flavour begins opening,
- sweetness improves,
- combustion stabilizes.

Final Third

- muddiness risk,
- declining combustion efficiency,
- possible draw heaviness.

What the Smoker Will Notice

The cigar may feel:

- smooth but muted,
- slow to develop,
- and flatter later in the smoke.

5.3 Root Cause

Typical causes include:

- recent RH reduction,
- dry-boxing,
- or wrapper moisture loss occurring faster than core moisture loss.

5.4 Corrective Actions

Recommended approach:

- allow short stabilization at lower RH,
- improve airflow,
- avoid excessive drying,
- and monitor delta convergence.

SECTION 6

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USING DELTAS TO PREDICT FLAVOUR TIMING

6.1 Early-Flavour Profile

Characteristics

- Cap ≈ Foot
- balanced ignition
- immediate aromatic release
- stable flavour progression

Professional Interpretation

The cigar is properly equalized.

6.2 Late-Bloom Profile

Characteristics

- Foot > Cap
- delayed sweetness
- stronger middle-third transition
- improving flavour progression

Professional Interpretation

The cigar has not fully equalized internally.

6.3 Front-Loaded Profile

Characteristics

- Cap > Foot
- rich early flavour
- softer combustion initially
- flattening later in the smoke

Professional Interpretation

The wrapper is responding faster than the filler core.

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Professional Perspective

Experienced sommeliers can intentionally shape:

- when flavours appear,
- how the cigar evolves,
- and how customers experience progression.

SECTION 7

DENSITY, BLENDER & VITOLA EFFECTS

7.1 Dense Cigars

Dense cigars generally:

- equalize more slowly,
- retain moisture longer,
- and show stronger cap lag effects.

Examples often include:

- My Father
- La Flor Dominicana

7.2 Delicate Cigars

Delicate cigars generally:

- respond quickly to RH change,
- show smaller tolerance windows,
- and become over-humidified easily.

Examples often include:

- many Cuban cigars,
- lighter Dominican puros.

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7.3 HOW VITOLA AFFECTS PEAK-FLAVOUR RH

7.3.1 Ring Gauge - The Dominant Variable

Narrow Vitolas (≤46 RG)

Examples:

- Lancero
- Corona
- Lonsdale

Characteristics:

- faster airflow,
- faster heat buildup,
- less filler buffering.

Professional implication:

Narrow cigars often prefer slightly higher RH.

Typical adjustment:
+0.5% to +1.0% RH

Thick Vitolas (≥54 RG)

Examples:

- Toro
- Gordo
- 6×60

Characteristics:

- slower airflow,
- cooler combustion,
- greater moisture buffering.

Professional implication:

Thick cigars often perform better at slightly lower RH.

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Typical adjustment:
−0.5% to −1.0% RH

7.3.2 Length - Secondary but Meaningful

Longer Cigars

- greater heat accumulation,
- longer combustion exposure,
- increased late-stage harshness risk if too dry.

Typical adjustment:
+0.25% to +0.5% RH

Shorter Cigars

- less thermal buildup,
- faster flavour delivery,
- lower harshness risk.

Typical adjustment:
Slightly lower RH tolerance.

7.3.3 Wrapper Dominance vs Filler Dominance

Thin Ring Gauge

- wrapper-forward combustion,
- higher aromatic sensitivity,
- more dryness-sensitive.

Thick Ring Gauge

- filler-dominant combustion,
- greater combustion buffering,
- stronger need for flavour clarity.

This explains why the same blend may feel:

- sharper in Lancero,

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- but muddier in Gordo

at the same RH.

7.3.4 Draw Resistance & Airflow Tuning

Vitola affects:

- airflow speed,
- oxygen availability,
- and combustion stability.

If airflow is too fast:

- combustion overheats,
- flavour sharpens,
- RH should increase slightly.

If airflow is too slow:

- combustion cools excessively,
- flavour mutes,
- RH should decrease slightly.

Peak RH compensates for combustion geometry.

Practical Adjustment Framework

Vitola	Typical Peak RH Adjustment
Lancero / Corona	+0.5–1.0%
Robusto (baseline)	0
Toro	−0.25–0.5%
Gordo / 6×60	−0.5–1.0%

Why This Matters for Lounges

Many lounges:

- store all vitolas identically,

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- serve all sizes at the same RH,
- and blame the cigar when performance suffers.

Professional lounges adjust preparation according to:

- geometry,
- combustion behaviour,
- and RH response.

Real-World Examples

Examples may include:

- My Father Lancero performing better at slightly higher RH than the Robusto,
- Warped Corto preferring lower RH than Maestro del Tiempo,
- Liga Privada Toro performing cleaner at lower RH than larger vitolas.

Same blend.
Different geometry.
Different combustion behaviour.

SECTION 8

APPLIED PROFESSIONAL SCENARIOS

Scenario 1 - Immediate Flavour Delivery

Customer Request

"I want the cigar to perform immediately."

Professional Response

- Select cigars with ≤1% delta,
- avoid recently adjusted cigars,
- prioritize fully equalized inventory.

Scenario 2 - Mid-Smoke Imbalance

Customer Complaint

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“The cigar became harsh halfway through.”

Professional Diagnosis

Possible cap lag or incomplete equalization.

Recommended Action

- explain moisture imbalance causally,
- recommend additional resting time before next smoke.

Scenario 3 - New Shipment Protocol

Professional Protocol

- Allow initial stabilization,
- measure progressively,
- avoid immediate service,
- and wait until deltas converge.

SECTION 9

EXAMINATION ALIGNMENT

Candidates must be able to:

- measure RH correctly,
- interpret cap-foot deltas,
- predict flavour evolution,
- and recommend corrective actions.

Common Examination Errors

Candidates often:

- ignore delta direction,
- focus only on RH magnitude,
- confuse wrapper behaviour with core behaviour,
- or assume all cigars equalize at the same speed.

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Professional analysis must remain:

- evidence-based,
- combustion-focused,
- and observational.

SECTION 10

CORE PROFESSIONAL PRINCIPLES

Candidates should be able to defend the following statements:

1. Cigars equalize axially rather than uniformly.
2. The cap is a strong proxy for internal core moisture.
3. Cap-foot delta predicts flavour timing.
4. Direction of imbalance matters as much as magnitude.
5. Balanced cigars usually announce themselves early in the smoke.

MODULE SUMMARY

This module establishes that:

- moisture gradients explain many inconsistent smoking experiences,
- cap-foot diagnostics allow pre-light prediction,
- flavour timing can be managed intentionally,
- and professional cigar advice becomes evidence-based.

With this framework mastered, the sommelier can confidently say:

“This cigar will open late - and I can explain why.”

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